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Palaeomagnetic Results from some Sydney Basin Igneous Rock Deposits

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ABSTRACT. The remanent magnetism of nine igneous deposits from the Sydney Basin has been measured and its stability investigated using alternating field and thermal demagnetisation techniques. Variations in the directions of magnetisation demonstrate that the deposits were formed during a number of discrete igneous episodes beginning as early as the Jurassic.

INTRODUCTION

Many small igneous bodies occur in the Sydney Basin. They form the main reserve of road metal in the Sydney area, and some of the larger deposits contain working quarries, a major source of crushed-rock also used in concrete manufacture. The reserves at the operating quarries are limited and it is important to investigate techniques which bear on the temporal and spatial relationships of the known deposits in order to provide guidelines for future exploration.

This paper presents the results of an investigation into the rock magnetism of some of the deposits, with a view to establishing a chronological classification. A sample of nine intrusives has been studied.

GEOLOGICAL SETTING

The mainly flat-lying sandstones and shales of the Sydney Basin are cut by many small igneous bodies (Crawford, 1973). These take the form of diatreme-like volcanic necks, dykes and sills. The volcanic necks (Hamilton, 1970) range in size from a few hectares to more than 16 ha, and in composition from volcanic breccia to massive basalt. The dykes, which are commonly basaltic in composition (Crawford, 1973) are less than 2 metres wide, and are not easily seen except on coastal sections.

PALAEOMAGNETIC PROCEDURES

Samples were collected to cover as wide an area, and hence in most cases as long a time of cooling, as possible, over the outcrop of each occurrence. Core samples were drilled with a 25 mm diameter rock drill, and oriented with a core-orienter and sun-compass using standard techniques (e.g. see Collinson *et al.*, 1967). The cores were sliced into specimens (22 m long) that were measured on a Digico Complete Results Rock Magnetometer. The coercive force spectrum and the related properties of stability of magnetisation were investigated with a Schonstedt alternating field demagnetizer (GSD-1).

Nine deposits were sampled, of which five were volcanic necks, two (Collaroy and Barrenjoey) were dykes, and two (Erskine Park and Kulnura) were sills (see Table 1). A number of specimens, one from each separately oriented drill-core sample (ranging in number from three to sixteen cores from each deposit) were stepwise demagnetised as listed in Table 1. Specimens from the remaining lengths of core were then treated in the field in which the pilot group showed minimum dispersion, shown asterisked, in Table 1 as maximum precision, using Fisher's (1953) best estimate of precision (k).

The blocking temperature temperature spectrum was investigated by treating selected specimens at incremental temperatures up to 700°C, in the

TABLE 1
PRECISION OF PILOT GROUPS AFTER CLEANING IN ALTERNATING FIELDS

Deposit	N	NRM	5 mT	10 mT	15 mT	20 mT	30 mT	40 mT	50 mT	60 mT
St Mary's	16	68.9	-	-	-	106*	-	-	-	-
Hornsby	6	27.7	29.6	62.6	97.9	177	148	266*	191	151
Collaroy	7	17.6	43.9	124	159*	119	13.0	13.0	2.0	-
Erskine Park	6	-	51.7	61.8	74.9*	74.2	70.7	74.0	63.8	57.0
Kulnura	4	24.0	196	280	217	242	202	971*	289	-
Peat's Ridge	7	82.4	227	315*	199	-	182	270	-	151
Mogo Hill	6	1.0	-	47.0*	-	9.3	8.8	-	-	-
Barrenjoey	7	-	18.5	118	200*	155	67.9	10.2	-	-
Minchinbury	3	215	142	488	-	258	2400*	619	-	-

N is the number of specimens used. The headings of the columns are the peak alternating fields used in demagnetising the specimens. The numbers in the columns represent Fisher's (1953) best estimate (k) of the precision of the group about the mean value.

TABLE 2
MEAN DIRECTIONS AND PRECISIONS AFTER OPTIMUM CLEANING

Deposit	Alternating Field						S.Pole Position		Thermal					
	Peak field (mT)	N	D	I	k	α_{95}	Lat.	Long.	Temp. (°C)	N	D	I	k	α_{95}
St Mary's	20	21	014	-81	960	3.3	51	144	500	5	028	-84	147	6.3
Hornsby	40	6	029	-62	191	4.8	66	92	400	3	034	-60	169	9.5
Collaroy	15	7	322	-75	159	4.8	53	180	600	4	255	-82	28	17.6
Erskine Park	15	12	184	-84	81	4.8	23	151	500	4	188	-86	78	10.4
Kulnura	40	8	032	-62	352	2.9	63	91	300	4	032	-59	239	5.9
Peat's Ridge	10	14	208	+63	333	2.1	66	98	300	4	163	+63	21	20.7
Mogo Hill	10	12	249	+84	41	6.8	37	138	200	3	305	+79	10	40.1
Barrenjoey	15	7	142	+74	200	4.2	53	182	600	3	148	+59	5.1	60.9
Minchinbury	30	8	320	-75	55	7.5	56	179	500	3	290	-76	146	10.2

N = number of specimens
D = declination
I = inclination
k = precision parameter (Fisher, 1953)
α_{95} = half-angle of cone of confidence at the 95% probability level

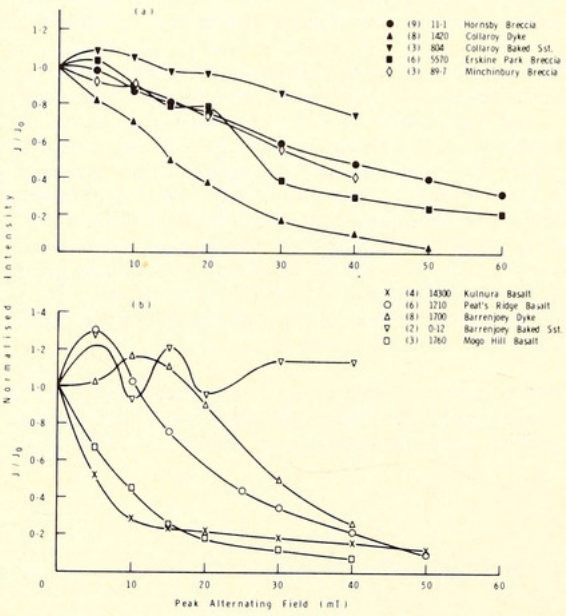


Figure 1. Normalised alternating magnetic field demagnetisation curves for deposits studied. (a) curves for Hornsby, Erskine Park and Minchinbury breccias, Collaroy dyke and Collaroy baked sediment (b) curves for Kulnura, Peat's Ridge and Mogo Hill basalts, Barrenjoey dyke and Barrenjoey baked sediment. Figures in brackets denote number of specimens used in deriving the curves. Unbracketed figures denote mean NRM intensities for same specimens in mA m^{-1} (gauss x 10^{-6})

Schonstedt (TSD-1) batch oven. The same minimum dispersion technique was used to obtain the optimum cleaning temperature.

RESULTS

The precision of directions about the mean for pilot groups for each formation (Fisher (1953) k), are given in Table 1 for peak alternating fields up to 60 mT. The fields yielding the maximum precision ranged from 10 mT for Peat's Ridge and Mogo Hill up to 40 mT for Hornsby and Kulnura. The maximum value, asterisked in the table, indicates the appropriate demagnetising field for the remaining material. The resultant mean directions, precision (k) and cone of the half-angle of confidence (Fisher's α_{95}) are shown in Table 2.

The magnetic characteristics of the deposits varied greatly. The initial directions from Mogo Hill were random, whereas from St Mary's, Peat's Ridge and Minchinbury they were tightly grouped. Alternating field cleaning improved the precision of directions of all formations. Specimen directions from all the igneous bodies studied grouped with low scatter after cleaning in peak alternating fields of either 10 or 15 mT. It seems likely that much of the initial scatter was caused by soft, viscous components.

The alternating magnetic field demagnetisation curves for each formation, shown in Fig. 1, reveal a wide range of coercivity spectra. The Kulnura and Mogo Hill basalts contain large low coercivity components (Fig. 1b), but they also have high initial intensities and therefore large components remaining in the high coercivity range. The baked sediments at Barrenjoey (Fig. 1b) and Collaroy (Fig. 1a) have very high coervicity components that are thought to be due to the presence of hematite. The increase in intensity for Peat's Ridge after cleaning in an alternating field of 5 mT is thought to be due to the preferential removal of a viscous component in the opposite sense to the reversely magnetised rock. The Hornsby, Erskine Park and Minchinbury breccia

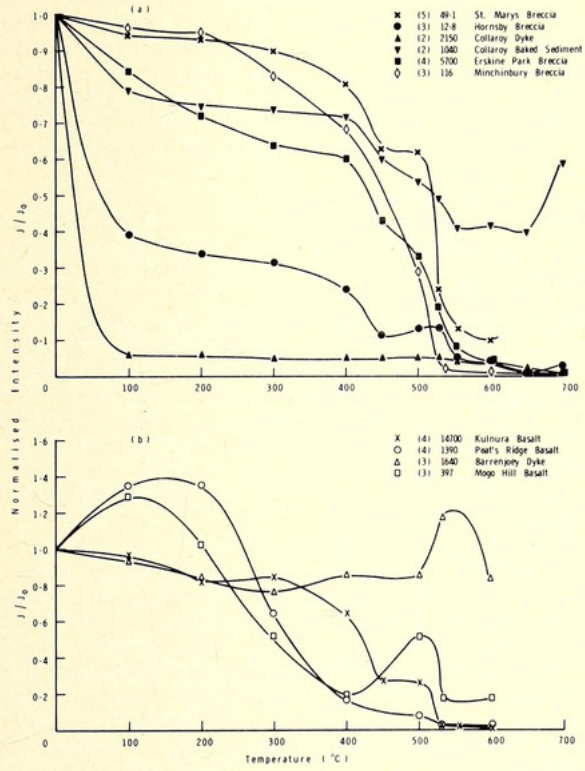


Figure 2. Normalised thermal demagnetisation curves for deposits studied (a) St Mary's, Hornsby, Erskine Park and Minchinbury breccias, Collaroy dyke and Collaroy baked sediment. (b) Kulnura, Peat's Ridge and Mogo Hill basalts and Barrenjoey dyke. Figures in brackets denote number of specimens from which curves were derived. Unbracketed figures denote mean NRM for same specimens in mA m^{-1}.

pipes have a wide range of coercivity spectra (Fig. 1a) which may be related to their explosive mode of formation.

Three to five specimens from each deposit were thermally demagnetised in steps (Fig. 2) up to 700°C for Barrenjoey, Mogo Hill and Collaroy baked sediment are thought to be due to chemical alteration during the heating. The increase of intensity for Peat's Ridge and Mogo Hill after cleaning to 100° and 200°C is probably due to the removal of low blocking-temperature components acquired in the present field direction; this opposes the ambient field direction at the time of formation.

The large blocking-temperature components in the range 400° to 550°C for St Mary's, Minchinbury and Erskine Park indicate that these breccias consolidated in temperatures in excess of 400°C, if we assume that the remanent magnetisation is of thermoremanent origin and acquired when the magnetic crystals cooled through these blocking temperatures. On the other hand, Hamilton *et al* (1970) suggest that, on the evidence of coal reflectance figures from fragments found in these breccias, the temperature of formation must have been less than 100°C. Two possible solutions to this inconsistency may be considered. Firstly, the temperature within the breccias may have varied from less than 100° to more than 400°C at the time of consolidation. Secondly, chemical alterations in the low temperature ranges may have caused the formation of magnetic minerals, most probably iron oxides that acquired a chemical remanence which itself had higher blocking temperatures.

DISCUSSION

The mean directions for the nine deposits are shown in Fig. 3. All are normally magnetised except Mogo Hill, Barrenjoey and Peat's Ridge, which are reversely magnetised. It is not possible to tell how much of secular variation has been averaged by the time spread of the samples, as the rate at which the acquisition of the magnetisation spread through the deposits is not known. Hence an unknown, but probably small, secular variation error is likely to be present in all the results. Bearing this in mind,

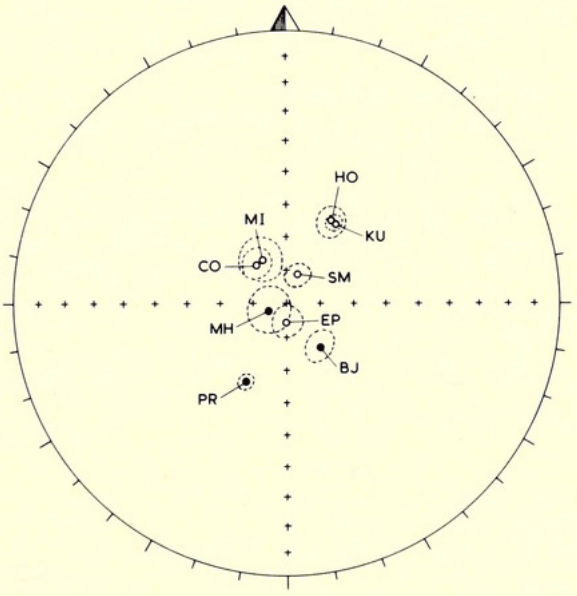


Figure 3. Mean directions of magnetisation for each deposit, using results from the alternating magnetic field that produced the highest precision, plotted on a Schmidt equal area net. Open (solid) circles are north-seeking directions on the upper (lower) hemisphere of the net. Areas outlined in broken lines represent the cones of confidence (Fisher, 1953). BJ = Barrenjoey dyke, CO = Collaroy dyke, EP = Erskine Park sill, HO = Hornsby volcanic neck, KU = Kulnura sill, MH = Mogo Hill volcanic neck, MI = Minchinbury volcanic neck, PR = Peat's Ridge volcanic neck, SM = St Mary's volcanic neck

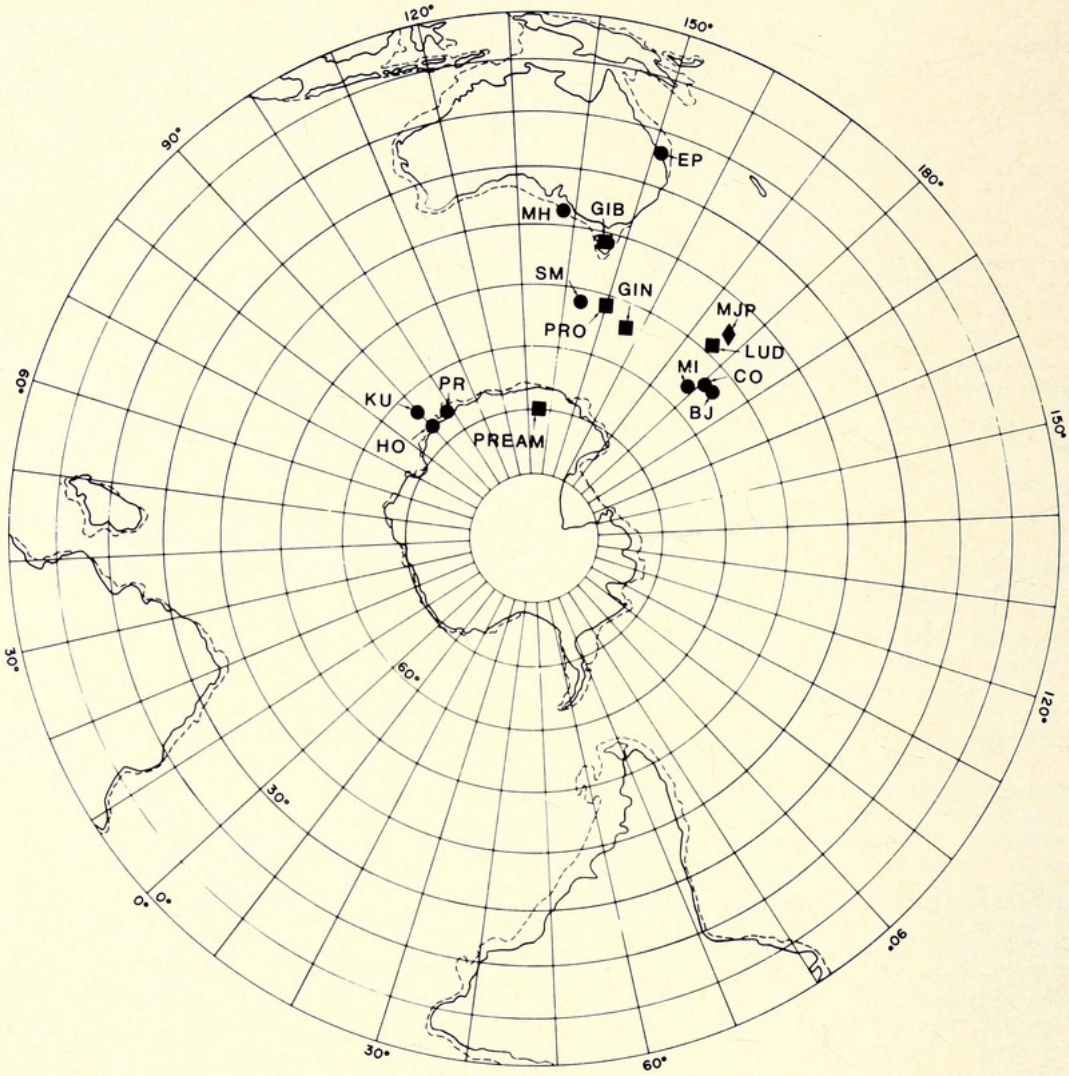


Figure 4. Equal area projection of the southern hemisphere of the earth, showing pole positions calculated from the directions shown in Fig. 3. Also shown are the pole positions for Gibraltar, GIB, Gingenbullen, GIN, and Prospect, PRO (Boesen *et al.*, 1961); Luddenham, LUD, and Peat's Ridge, PREAM (Manwaring, 1963); and the mean Jurassic role (MJP) of Schmidt (1976). MJP now supercedes GIB, GIN and PRO.

the mean directions fall into two distinct groups (Fig. 4). Hornsby, Kulnura and Peat's Ridge form one group, and contain a Tertiary field direction. The fact that the Peat's Ridge direction is reversed indicates that the deposit cannot be exactly contemporaneous with Kulnura and Hornsby but, on the present palaeomagnetic evidence, it is not possible to say which is the older. The other group consists of Collaroy, Minchinbury and Barrenjoey, consistent with a Jurassic age. Pole positions for the deposits at Mogo Hill, Erskine Park and St Mary's are scattered. The intrusions into the Sydney Basin came in several pulses, the first of which was not long after the deposition of the basin. The radiogenic dating of one or more of these palaeomagnetically 'older' intrusions would be of great interest.

Pole positions for other intrusions into the Sydney Basin, of which the directions had been previously determined, (Boesen *et al.*, 1961; Manwaring, 1963) are also shown in Fig. 4.

Gilbraltar, Prospect and Gingenbullen have been radiogenically dated (Evernden and Richards, 1962) and appear to have been intruded during the Jurassic. Recent work by Schmidt (1976) demonstrates that the pole positions from this earlier palaeomagnetic work result from directions in which secondary components have not been entirely eliminated, and his mean Jurassic pole position (MJP), from rocks subjected to more severe laboratory treatment, is also shown in Fig. 4.

The results of this preliminary study show that there are significant differences in remanent magnetisation directions between different intrusions into the Sydney Basin. They are consistent with a hypothesis that the intrusions into the Sydney Basin came in several pulses (Manwaring, 1963) that can be differentiated, in some cases, by differences in their magnetisation directions and the pole positions calculated from them. This technique may 'll provide a simple method of dividing such intrusions into broad age groups.

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